Venous Thromboembolism (VTE) Management In Oncology Patients: From Risk Assessment to Personalized Anticoagulation

Ascension

Listening to you, caring for you.®

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No financial disclosure

Objectives

- ✓ Understand the epidemiology and clinical significance of VTE in cancer patients
- ✓ Apply risk assessment tools, like Khorana score, to guide VTE prophylaxis decisions.
- ✓ Interpret landmark data (Hokusai, SELECT-D, CARAVAGGIO, CANVAS)
- ✓ Implement current ASCO (2023), ITAC (2022) and NCCN (2025) guideline recommendations.

Abbreviations

- **ASCO:** American Society of Clinical Oncology
- **CAT**: Cancer-Associated Thrombosis
- **CRNMB:** Clinically Relevant Non major Bleeding
- **DOACs**: Direct Oral Anticoagulants
- **DVT**: Deep Vein Thrombosis
- **DSMB**: Data Safety Monitoring Board
- **GI**: Gastrointestinal/**GU**: Genitourinary
- **IMiDs**: Immunomodulatory drugs
- **ITAC:** International Initiative on Thrombosis and Cancer
- **LMWH**: Low Molecular Weight Heparin
- **NCCN**: National Comprehensive Cancer Network
- **PE**: Pulmonary Embolism
- **TF:** Tumor-derived tissue factor
- **VTE** : Venous Thromboembolism

Armand Trousseau described the association between cancer and thrombosis 1865

Armand Trousseau described the association between cancer and thrombosis in 1865

- Trousseau identified migratory thrombophlebitis as a predisposing factor of malignancy and later diagnosed the syndrome in himself before succumbing to gastric cancer.
- Trousseau's syndrome: a recurrent thrombosis in superficial veins and unusual sites (chest wall, amrs) are strongly associated with pancreatic, gastric and lung cancer.

Epidemiology & Clinical Significance

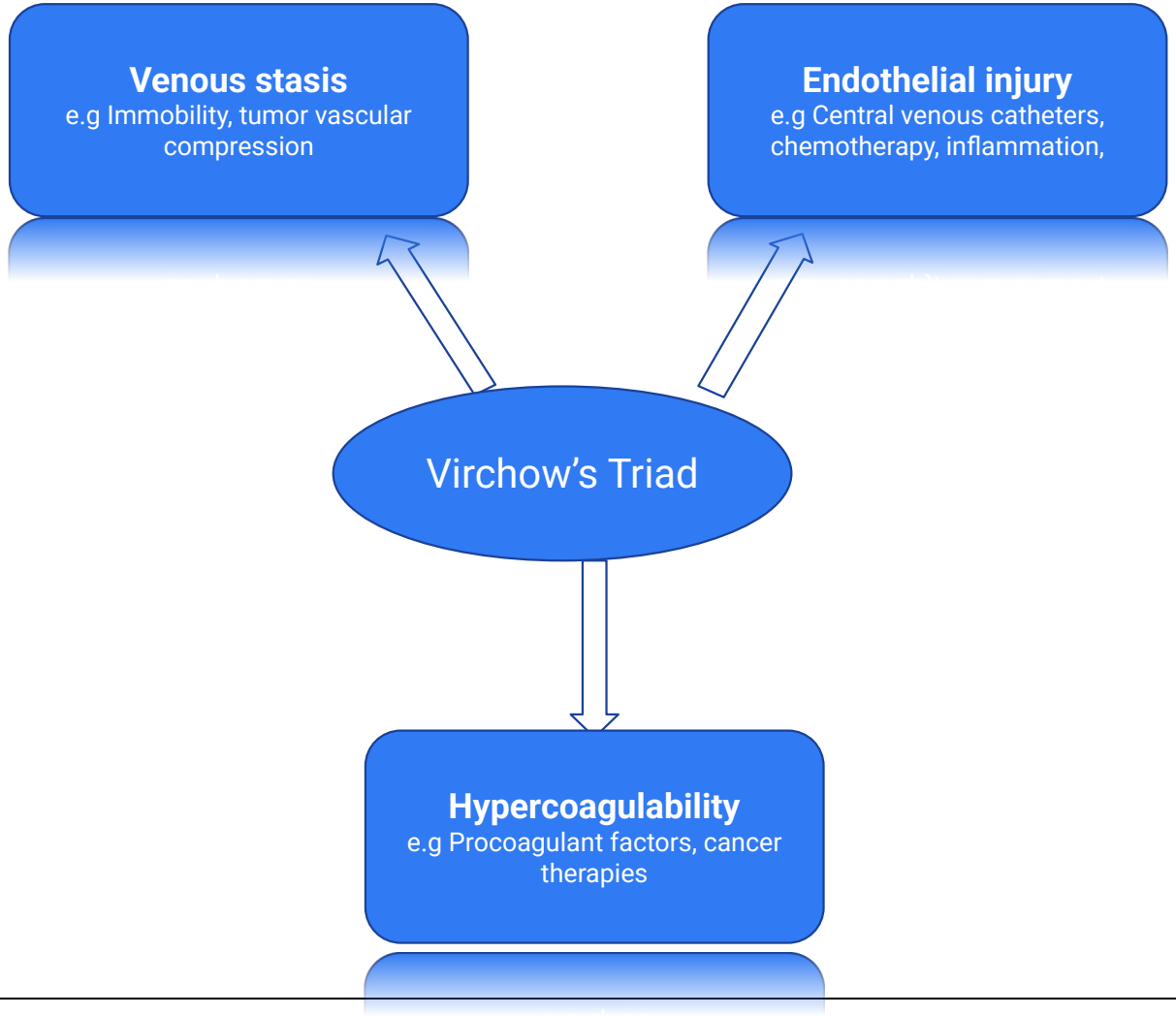
MORTALITY IMPACT

VTE is the second leading cause of death in ambulatory cancer patients receiving chemotherapy

- **Incidence**: Affects 4 - 20% of patients; highest rates observed in pancreatic, gastric and brain cancers.
- **High-Risk Periods:** First 3-6 months of chemotherapy, perioperative period, and during hospitalization.
- **Treatment Impact**: Results in treatment delays, dose reductions, therapy interruptions and reduced quality of life
- **Healthcare Burden:** Associated with increase healthcare utilization and hospital readmission rates.

Gandarilla O, Taiwo E, Parikh MA, Astrow AB, Frishman WH, Peterson SJ. Cancer-associated thromboembolic disease and the evolution of management. *Cardiol Rev*. Published online November 22, 2025. doi:10.1097/CRD.0000000000001119

Pathophysiology: Virchow's Triad in Cancer



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Mosarla RC, Vaduganathan M, Qamar A, Moslehi J, Piazza G, Giugliano RP. Anticoagulation strategies in patients with cancer. *J Am Coll Cardiol*. 2019;73(11):1336-1349. doi:10.1016/j.jacc.2019.01.017

Tumor-Derived Procoagulants

- Tissue factor overexpression activates the extrinsic coagulation pathway
- TF-positive extracellular vesicles shed from tumors bind to factor VII to generate thrombin
- Cancer procoagulant directly activates factor X (independent of factor VII)
- Brain tumors activates platelets via CLEC-2 receptor

Cancer Therapy Induced Procoagulation

Chemotherapy:

- Direct endothelial injury
- Platinum agents (cisplatin, carboplatin) are particularly thrombogenic
- Tumor lysis releases procoagulants and cytokines
- Hepatotoxicity decreases natural anticoagulants (protein C, protein S, antithrombin)
- Reduces thrombomodulin + increase factor VIII → acquired protein C resistance

Targeted Therapies

VEGF (Vascular Endothelial Growth Factor) Inhibitors (bevacizumab)

- Block VEGF-mediated endothelial protection
- endothelial dysfunction, hypertension, arterial and venous thrombosis

BCR-ABL Inhibitors (ponatinib)

- BBW for accelerated atherosclerosis

Immune Checkpoint Inhibitors:

- Activate inflammatory pathways →systemic inflammation, vasculitis

Hormonal Oral Therapies:

- Tamoxifen: estrogenic effects on coagulation
- Thalidomide/lenalidomide especially with dexamethasone

Radiation Therapy:

- Acute and chronic endothelial injury that persists years after treatment

Knowledge Check

Which of the following statements regarding the clinical significance of VTE in oncology is TRUE?

A. VTE is the leading cause of death in cancer patients

B. The highest rates of VTE are observed in breast and prostate cancers

C. VTE is the second leading cause of death in ambulatory cancer patients receiving chemotherapy.

D. High risk periods for VTE are usually the first month of diagnosis.

Knowledge Check

Which of the following statements regarding the clinical significance of VTE in oncology is TRUE?

A. VTE is the leading cause of death in cancer patients

B. The highest rates of VTE are observed in breast and prostate cancers

C. **VTE is the second leading cause of death in ambulatory cancer patients receiving chemotherapy.**

D. High risk periods for VTE are usually the first month of diagnosis.

Primary Prophylaxis

Risk Stratification: Khorana Score for VTE Risk

Patient Characteristics & Laboratory Parameters	Points
Site of Cancer: Very High Risk Gastric, Pancreas	2
Site of Cancer: High Risk Lung, Lymphoma, Gynecologic, Bladder, Testicular	1
Pre-chemotherapy Platelet Count $\geq 350 \times 10^9$ /L	1
Hemoglobin < 10 g/dL or use or Erythropoiesis-Stimulating Agents	1
Pre-chemotherapy Leukocyte Count $> 11 \times 10^9$ /L	1
Body Mass Index (BMI) ≥ 35 kg/m^2	1

Risk Interpretation:

0 points = **Low** → No routine prophylaxis

1-2 points = **Intermediate** → Consider prophylaxis; discuss with patient

≥3 points = **High** → Prophylaxis recommended (ASCO)

Note: *ASCO recommends prophylaxis consideration at score ≥2; NCCN categorizes ≥3 as high risk. The Khorana score is validated for ambulatory patients initiating chemotherapy, not for hospitalized patients or those with established VTE*

National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines): Cancer-Associated Venous Thromboembolic Disease. Version 2.2025. Published May 22, 2025. Accessed January 10, 2026. https://www.nccn.org/professionals/physician_gls/pdf/vte.pdf

Clinical Management: Primary Prophylaxis Recommendations

Setting	Agent	Duration
Hospitalized	LMWH preferred (UFH if CrCl <30 mL/min) Enoxaparin 40 mg SQ daily	During admission, extended prophylaxis not recommended
Ambulatory high-risk (Khorana ≥2)	Apixaban 2.5 mg BID or Rivaroxaban 10 mg daily	Up to 6 months during chemotherapy
Post-surgical (abdominal/pelvic)	LMWH Enoxaparin 40 mg SQ daily	Extended 4 weeks post-discharge
Multiple myeloma on IMiDs (e.g lenalidomide, thalidomide)	Aspirin, LMWH, or warfarin based on risk per IMWG guidelines	Duration of IMiD therapy

⚠️ Caution with DOACS in GI/GU cancers, assess mucosal bleeding risk, use LMWH over DOACs

National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines): Cancer-Associated Venous Thromboembolic Disease. Version 2.2025. Published May 22, 2025. Accessed January 10, 2026. https://www.nccn.org/professionals/physician_gls/pdf/vte.pdf

Knowledge Check

According to the Khorana Score, which of the following patients would be considered “High Risk” for VTE ?

A. A patient with lung cancer (1pt) with a BMI of 25 kg/m² (0 pts).

B. A patient with pancreatic cancer (2pts) and a pre-chemotherapy platelet count of 400 x 10^9/L (1 pt)

C. A patient with gastric cancer (2pt) and a hemoglobin of 12 g/dL (0pts)

D. A patient with lymphoma (1 pt) and a leukocyte count of 10 x 10^9/L (0 pts).

Knowledge Check

According to the Khorana Score, which of the following patients would be considered “High Risk” for VTE?

A. A patient with lung cancer (1pt) with a BMI of 25 kg/m² (0 pts).

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C. A patient with gastric cancer (2pt) and a hemoglobin of 12 g/dL (0pts)

D. A patient with lymphoma (1 pt) and a leukocyte count of 10 x 10^9/L (0 pts).

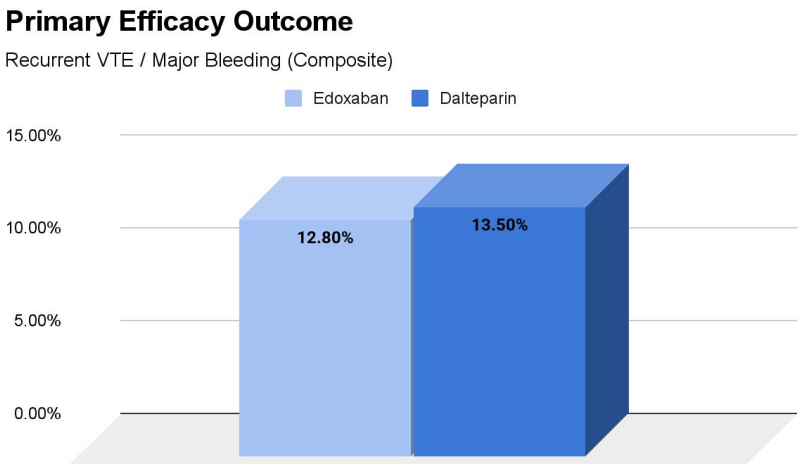
LANDMARK CLINICAL TRIALS

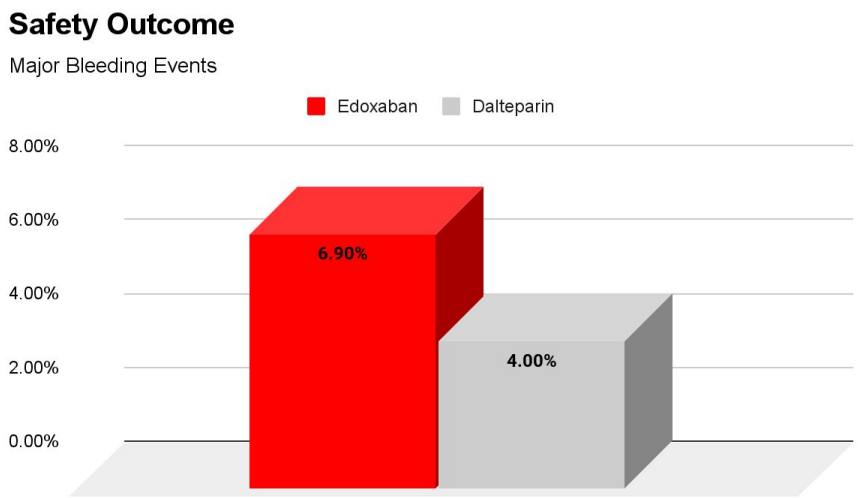
Hokusai VTE Cancer (2018) Overview

Goal: To evaluate if oral Edoxaban is non-inferior to Dalteparin for the treatment of cancer-associated VTE for 12 months

- **Design**: Multinational, randomized, open-label, non-inferiority trial (N= 1,050)
- **Patient Population:**
  - **Cancer types:** Colorectal, lung, breast, pancreatic, gastric and hematologic malignancies
  - **Inclusion:** Active cancer (diagnosed/treated within 6 months or metastatic), or cancer diagnosis within 2 years, acute symptomatic or incidental proximal DVT or PE
  - **Exclusion criteria:** CrCl <30 mL/min, platelets <50k, high bleeding risk (active bleeding, CNS tumor or need for DAPT).
  - **Primary outcome:** First recurrent VTE or major bleeding during 12 month period

Hokusai VTE Cancer (2018)





Key Takeaways

- ❑ Edoxaban met non-inferiority for the composite endpoint of recurrent VTE and major bleeding (12.8% vs 13.5%, HR 0.97, 95% CI 0.70-1.36)
- ❑ Treatment duration was for 12- months which addressed evidence gap beyond 6 months
- ❑ Clinical relevance: Lower VTE recurrence but higher major bleeding driven by GI malignancies
- ❑ Edoxaban is a viable option but requires 5-day LMWH lead-in; avoid in GI malignancies

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Raskob GE, van Es N, Verhamme P, et al. Edoxaban for the treatment of cancer-associated venous thromboembolism. *N Engl J Med*. 2018;378(7):615-624. doi:10.1056/NEJMoa1711948

Critical Appraisal- Hokusai VTE Cancer (2018)

Strengths:

- Study design: Randomized, large, multicenter design (1,050 patients, 114 sites)
- 12- month treatment duration addressed existing evidence gap for prolonged anticoagulation beyond 6 months.
- Composite primary endpoint has a balance between efficacy and safety

Limitations:

- Open-label design introduces selection bias
- Composite endpoints overshadows opposing effect: fewer VTE recurrences but the incidence of significant major bleeding was higher with edoxaban (6.9% vs 4.0%, HR 1.77, p=0.04)
- Treatment duration was imbalanced 211 days with edoxaban vs 184 days with dalteparin
- About 23% of the subjects no longer had active cancer at enrollment which may limit applicability to our sickest patients.

Clinical Implication: Edoxaban works but not appropriate in GI cancer, LMWH lead-in is a barrier.

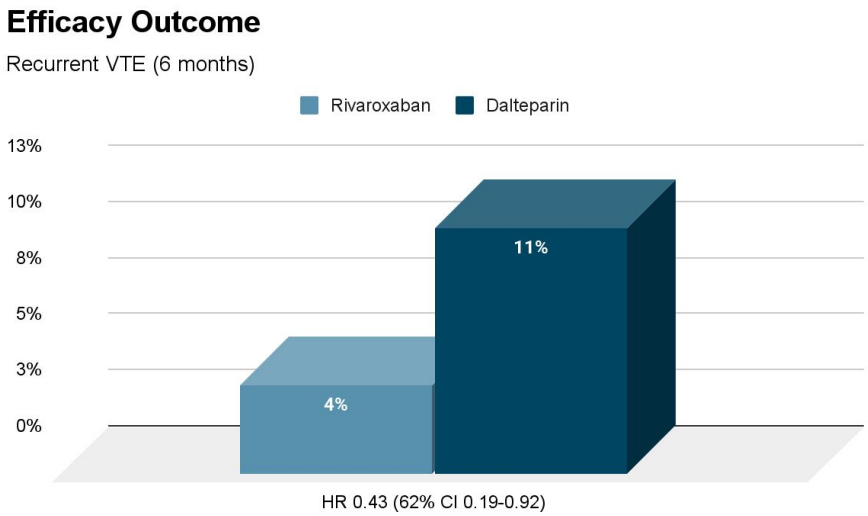
SELECT-D (2018) Overview

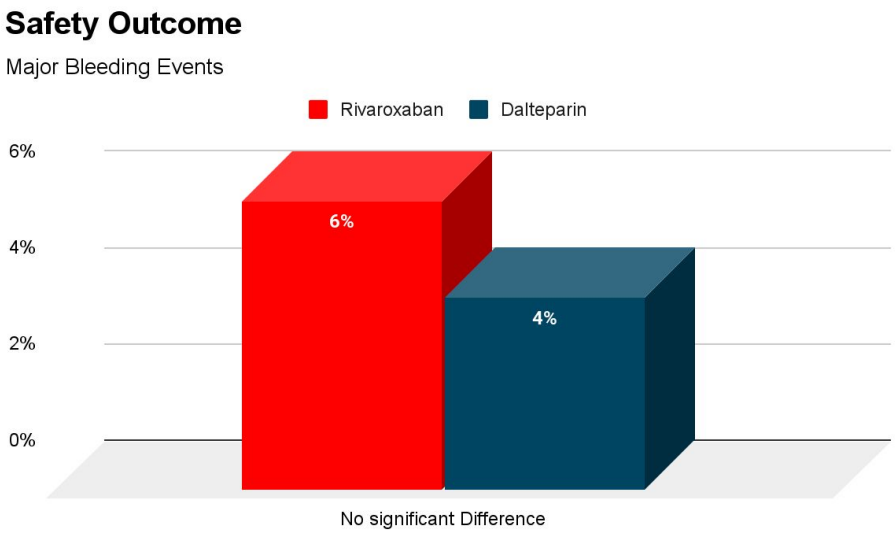
Goal: To evaluate if oral Rivaroxaban is a safe and effective alternative to Dalteparin for cancer associated VTE

Design: Multinational, randomized, open-label, pilot trial in the UK (N= 406)

- **Patient Population:**
  - **Cancer types:** Colorectal, lung, breast and pancreatic (Esophageal and gastroesophageal cancers were initially included but excluded)
  - **Inclusion:** Active cancer with symptomatic PE, incidental PE, or symptomatic proximal DVT, ECOG PS 0-2
  - **Exclusion criteria:** CrCl <30 mL/min, platelets <100k, active bleeding, strong CYP3A4 or P-gp inhibitors or inducers
  - **Primary outcome:** VTE recurrence over 6 months

SELECT-D (2018)





Key Takeaways

- ❑ Rivaroxaban showed significant VTE reduction (4% vs 11%, HR 0.43)
- ❑ Exclusion of upper GI cancers mid-trial due to the occurrence of significant major bleed (36% rivaroxaban vs 11% dalteparin).
- ❑ CRNMB significantly higher with rivaroxaban
- ❑ Avoid in upper GI malignancies due to increased bleeding risk.

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Young AM, Marshall A, Thirlwall J, et al. Comparison of an oral factor Xa inhibitor with low molecular weight heparin in patients with cancer with venous thromboembolism: results of a randomized trial (SELECT-D). *J Clin Oncol*. 2018;36(20):2017-2023. doi:10.1200/JCO.2018.78.8034

Critical Appraisal- SELECT-D (2018)

Strengths:

- Trial showed statistically significant VTE reduction benefit with rivaroxaban (HR 0.43, 95% CI 0.19-0.99)
- Pragmatic regimen mirrors real world clinical setting
- Data and Safety Monitoring Board excluded upper GI cancers mid-trial due to a higher incidence of major bleeding in esophageal/gastric cancer with rivaroxaban (36% vs 11%)

Limitations:

- Pilot study design with N= 406 was not powered for concrete conclusion.
- Open label design introduces selection bias
- 3.76 fold increase in CRNMB (13% vs 4%) raises a safety concern
- Subjects were only from the UK, limits external validity.
- Adjusting protocols mid trial to exclude GI patients limits external validity

CARAVAGGIO (2020) Overview

Goal: To determine if Apixaban is non-inferior to Dalteparin for the treatment of cancer associated VTE without increasing bleeding risk

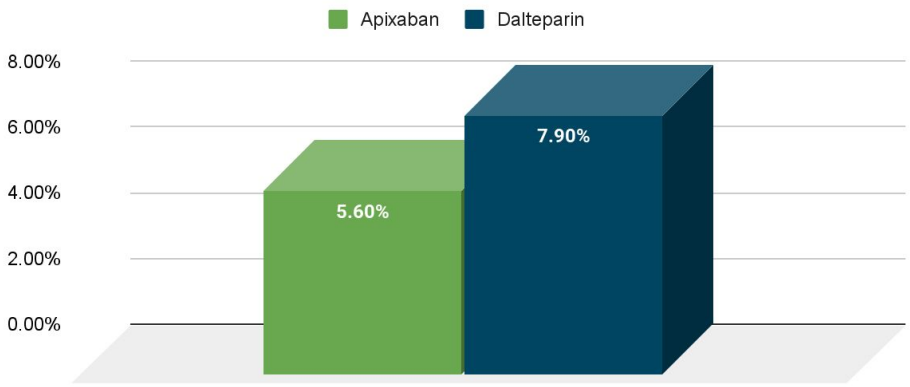
Design: Multinational, randomized, open-label, non-inferiority trial (N= 1,170)

- **Patient Population:**
  - **Cancer types:** 33% of GI cancers, lung, breast and GU cancers
  - **Inclusion:** Acute symptomatic or incidental proximal DVT or PE, active cancer (diagnosed within 6 months or receiving treatment) or cancer diagnosis within 2 years
  - **Exclusion criteria:** Brain tumors or known cerebral metases, acute leukemia, CrCl <30mL/min, platelets <75k
  - **Primary outcome:** Recurrent VTE, major bleeding and GI specific major bleeding

CARAVAGGIO (2020)

Primary Efficacy Outcome

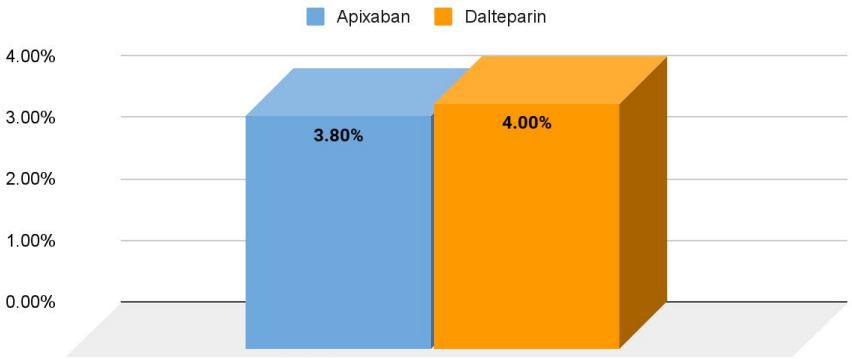
Recurrent VTE



Non-inferior (p0.001)

Safety Outcome

Major Bleeding Events



No Significant Difference (P=0.60)

Key Takeaways

- ❑ Apixaban significantly reduced VTE recurrence compared to Dalteparin
- ❑ No increased major bleeding observed, including in GI cancers
- ❑ Excluded brain tumors, brain metastases, acute leukemia
- ❑ Apixaban is preferred in GI/GU malignancies

Ascension

Agnelli G, Becattini C, Meyer G, et al. Apixaban for the treatment of venous thromboembolism associated with cancer. *N Engl J Med*. 2020;382(17):1599-1607. doi:10.1056/NEJMoa1915103

Critical Appraisal- CARAVAGGIO (2020)

Strengths:

- Large apixaban trial (N=1,170)
- Minimal GI bleeding and best safety profile compared to the other trials
- Oral monotherapy with apixaban
- Subgroup analysis was comprehensive with consistent results across the various cancer types

Limitations:

- Excluded brain tumors, brain metastases and acute leukemia (limits generalizability)
- Open label design may influence subjective outcomes
- Study duration was only 6 months, not powered for superiority (p=0.09)
- No direct comparison between DOACs

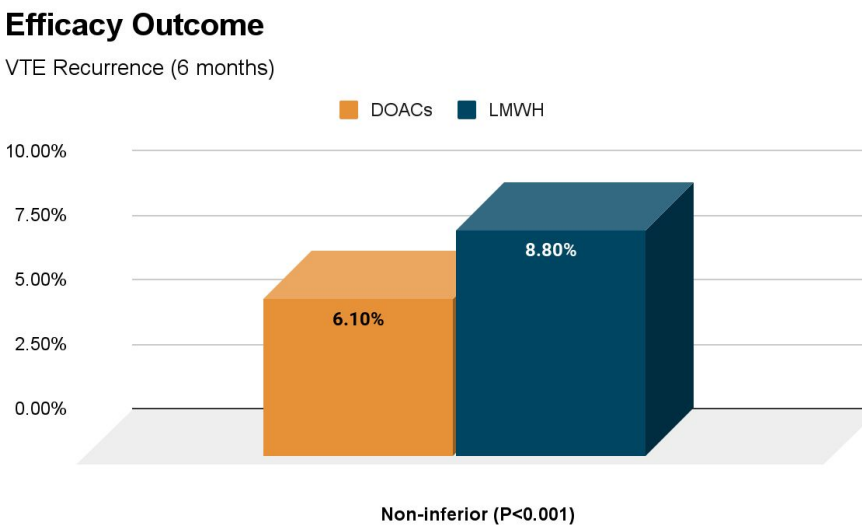
CANVAS (2022/2023) Overview

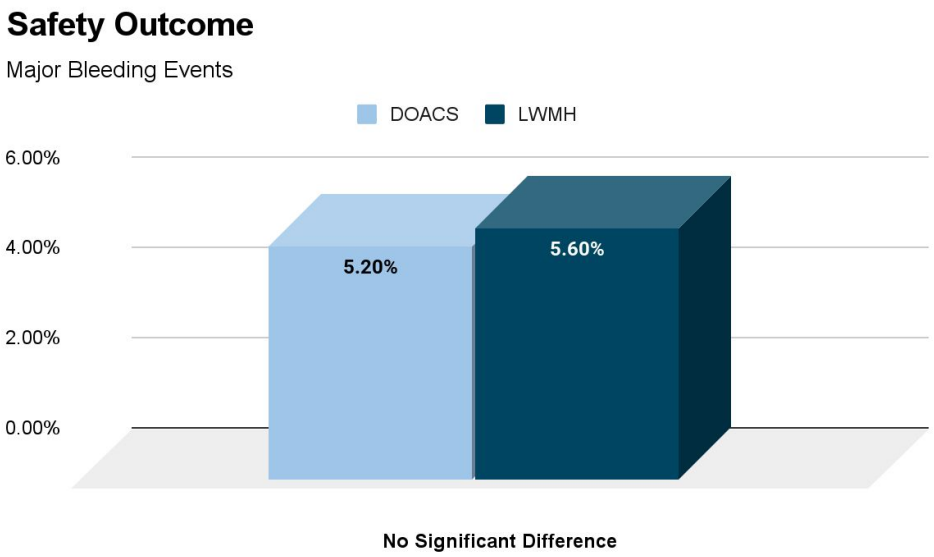
Goal: To evaluate the effectiveness of investigator's choice DOACs compared to LMWH for preventing recurrent VTE in cancer patients in real-world clinical setting

Design: Multinational, randomized, open-label, pragmatic, non-inferiority trial (N= 671)

- **Patient Population:**
  - **Cancer types:** Any invasive solid tumors, lymphoma, multiple myeloma or chronic lymphocytic leukemia.
  - **Inclusion:** New diagnosis of VTE within 30 days of enrollment
  - **Exclusion criteria:** Active bleeding, severe renal impairment
  - **Primary outcome:** Recurrent VTE at 6 months
  - **Safety Outcome:** Major bleeding

CANVAS (2022/2023)





Key Takeaways:

- ❑ Validated DOACs use in broader real-world population
- ❑ 98.5% adherence with DOACS vs 91.7% with LMWH
- ❑ Study discontinued early due to COVID-19
- ❑ DOACs are effective and safe in fragile populations excluded from previous trials

Schrag D, Uno H, Rosovsky R, et al. Direct oral anticoagulants vs low-molecular-weight heparin and recurrent VTE in patients with cancer: a randomized clinical trial. *JAMA*. 2023;329(22):1924-1933. doi:10.1001/jama.2023.7843

Critical Appraisal- CANVAS (2022/2023)

Strength:

- Pragmatic study mirrors real-world clinical setting
- More broad inclusion criteria, including brain metastases, impaired performance status and poor renal function.
- Higher treatment adherence with DOACs 98.5% vs 91.7% with LMWH
- Multiple oncology sites in the USA

Limitations:

- Open label design introduces detection bias
- Difficult to differentiate between fatal VTE and death from cancer
- COVID-19 early termination
- Minority ethnic groups underrepresented (83% caucasian, 11% black and 5% hispanic)

Trial Comparison Table

Trial	Number of subjects	DOAC	Comparator	Population
Hokusai	1,050	Edoxaban 60 mg QD	Dalteparin	Symptomatic/incidental VTE
SELECT-D	406	Rivaroxaban 15 → 20 mg QD	Dalteparin	PE or proximal DVT
CARAVAGGIO	1,170	Apixaban 10 mg →5 mg BID	Dalteparin	Symptomatic/incidental VTE (excluded brain metastases)
CANVAS	671	Investigator choice	LMWH(Enoxaparin, Dalteparin, Fondaparinux)	Broad inclusion (brain metastases included)

What did we Learn?

Trial	VTE Recurrence	Major Bleeding	Key Signal
Hokusai (Edoxaban vs Dalteparin)	7.9 % vs 11.3%	6.9% vs 4.0 %	↑ GI bleeding
SELECT-D (Rivaroxaban vs Dalteparin)	4% vs 11%	6% vs 4%	↑ Upper GI bleeding → excluded mid-trial
CARAVAGGIO (Apixaban vs Dalteparin)	5.6% vs 7.9%	3.8% vs 4.0%	**No ↑ GI bleeding**
CANVAS (Investigator's choice of DOAC)	6.1% vs 8.8%	5.2 % vs 5.6%	**No ↑ GI bleeding**

Knowledge Check

Which of the following landmark trial demonstrated that Apixaban was non-inferior to Dalteparin without an increased risk of major gastrointestinal (GI) bleeding?

A. Hokusai VTE Cancer

B. SELECT-D

C. CARAVAGGIO

D. CANVAS

Knowledge Check

Which of the following landmark trial demonstrated that Apixaban was non-inferior to Dalteparin without an increased risk of major gastrointestinal (GI) bleeding?

A. Hokusai VTE Cancer

B. SELECT-D

C. **CARAVAGGIO**

D. CANVAS

CURRENT CLINICAL GUIDELINES TREATMENT RECOMMENDATIONS

ASCO vs ITAC vs NCCN

Clinical Domain	ASCO 2023	ITAC 2022	NCCN 2025
Firstline	DOAC = LMWH	DOAC preferred	DOAC = LMWH
Minimum therapy duration	6 months	6 months	3 months
Extended therapy	Indefinite if cancer is still active	Reassess after 6 months	Consider apixaban 2.5 mg BID after 6 months

All guidelines emphasize shared decision making and individualized to patient preference

Choosing Between DOACs and LMWH

Clinical Scenario	Preferred Agent	Key Rationale
Most ambulatory patients	**DOAC** (apixaban or rivaroxaban)	Oral convenience, 34% VTE reduction vs LMWH
GI/GU cancer	**Apixaban**	No increased major GI bleeding (CARAVAGGIO)
Unresected GI tumor with mucosal lesion	**LMWH**	Avoid direct luminal anticoagulant exposure
CrCl 30 mL/min	**LMWH**	DOACs not recommended; monitor anti-Xa
Significant drug to drug interactions (CYP3A4 inhibitor)	**LMWH**	Avoids CYP/P-gp metabolism
Thrombocytopenia (< 50,000/microliter)	**LMWH (dose adjusted)**	Can titrate; hold DOACs

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Mousavi A, Shojaei S, Rahmati S, et al. Direct oral anticoagulants versus low-molecular-weight heparin in patients with cancer-associated thrombosis: a meta-analysis of randomized controlled trials. *CVP*. 2024. doi:10.1016/j.cpcardiol.2024.102919
Streiff MB, Holmstrom B, Angelini D, et al. Cancer-associated venous thromboembolic disease, version 2.2024, NCCN clinical practice guidelines in oncology. J Natl Compr Canc Netw. 2024;22(7):483-506. doi:10.6004/jnccn.2024.0046

Dosing Regimens for Cancer VTE

ANTICOAGULANT	INITIAL DOSE	MAINTENANCE DOSE	RENAL & SPECIAL ADJUSTMENT
Apixaban (Direct FXA Inhibitor)	**10 mg BID** oral for the first 7 days	**5 mg BID** oral thereafter	**Renal Caution:** ● standard dose if CrCl > 30 mL/min ● consider 2.5 mg BID if CrCl < 30 or meeting reduction criteria (Age ≥ 80, Wt ≤ 60kg, Cr ≥ 1.5)
Rivaroxaban (Direct FXA Inhibitor)	**15 mg BID** oral for the first 21 days (with food)	**20 mg Daily** (with food)	**Dose Reduction:** ● Reduce to 15 mg Daily if CrCl 30-49 mL/min ● Avoid if CrCl < 30mL/min
Edoxaban (Direct FXA Inhibitor)	**LMWH Lead-in** (Enoxaparin 1mg/kg BID x 5 days minimum)	**60 mg Daily** (Start 12 hours after lead-in)	**Reduce to 30 mg Daily if**: ● CrCl 30-50 mL/min ● Weight if ≤ 60kg ● Concomitant P-gp inhibitors
Dalteparin (LMWH)	**200 IU/kg Daily** For the first 30 days	**150 IU/kg Daily** Continuous Thereafter	**Renal/Platelet:** ● Monitor anti-Xa levels if CrCl < 30 mL/min ● Hold if platelets <50k or dose reduce

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Lexicomp. Dalteparin, Apixaban, Rivaroxaban, Edoxaban. In: *Lexicomp Online*. Wolters Kluwer Health, Inc; 2026. Updated January 5, 2026. Accessed January 11, 2026. https://online.lexi.com

Knowledge Check

In a patient with an unresected gastrointestinal tumor and high mucosal bleeding risk, which anticoagulant is generally preferred by clinical guidelines?

A. Rivaroxaban

B. Low molecular weight heparin (Enoxaparin 1mg/kg/bid)

C. Edoxaban

D. Apixaban

Knowledge Check

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B. Low molecular weight heparin (Enoxaparin 1mg/kg/bid)

C. Edoxaban

D. Apixaban

Clinical Management: Duration of Therapy

Treatment Timeline & Decision Nodes

Initial Diagnosis (Month 0)
Start therapeutic anticoagulation (DOAC or LMWH). Focus of acute stability.

↓

6-Month Milestone
Critical Decision:
- Is cancer active? if Yes continue
- If cancer resolve, consider stopping

↓

Extended Therapy (> 6 Months)
Continue if: Metastatic disease or ongoing chemotherapy.
Consider lower dose for stable patients with lower risk

STANDARD RECOMMENDATION
Treat for a Minimum of 6 Months

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Farge D, Frere C, Connors JM, et al. 2022 international clinical practice guidelines for the treatment and prophylaxis of venous thromboembolism in patients with cancer, including patients with COVID-19. *Lancet Oncol*. 2022;23(7):e334-e347. doi:10.1016/S1470-2045(22)00160-7
National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines): Cancer-Associated Venous Thromboembolic Disease. Version 2.2025. Published May 22, 2025. Accessed January 10, 2026. https://www.nccn.org/professionals/physician_gls/pdf/vte.pdf
Key NS, Khorana AA, Kuderer NM, et al. Venous thromboembolism prophylaxis and treatment in patients with cancer: ASCO guideline update. *J Clin Oncol*. 2023;41(16):3063-3071. doi:10.1200/JCO.23.00294

Incidental VTE in Cancer Patients

Definition: VTE detected on imaging performed for other indications (e.g staging CT scans)

Prevalence: Found in 1-5% of staging CT scans in cancer patients

Clinical Significance:

- Associated with similar rates of recurrent VTE and mortality as symptomatic VTE
- ASCO, ITAC and NCCN recommend treating incidental VTE the same as symptomatic VTE

Management Approach:

- Confirm findings with dedicated imaging if uncertain
- initiate therapeutic anticoagulation

Main Takeaway: Do not dismiss incidental findings, they carry some prognostic significance and need treatment

CLINICAL CASE STUDIES

Clinical Case Presentation: VTE Prophylaxis

EP is a 58 years old male about to start FOLFIRINOX for Metastatic Pancreatic Adenocarcinoma with a ECOG PS: 1

Pertinent Laboratory Data:
CrCl: 65 mL/min
Platelet: 285 x 10^9/L
Hemoglobin: 9.1 g/dL

Bleeding History: No active GI bleeding or mucosal lesions

Patient preference: prefers oral therapy

Does the patient need prophylaxis? What is the appropriate agent?

Risk Stratification: Khorana Score for VTE Risk

Patient Characteristics & Laboratory Parameters	Points
Site of Cancer: Very High Risk Gastric, Pancreas	2
Site of Cancer: High Risk Lung, Lymphoma, Gynecologic, Bladder, Testicular	1
Pre-chemotherapy Platelet Count $\geq 350 \times 10^9$ /L	1
Hemoglobin < 10 g/dL or use or Erythropoiesis-Stimulating Agents	1
Pre-chemotherapy Leukocyte Count $> 11 \times 10^9$ /L	1
Body Mass Index (BMI) ≥ 35 kg/m^2	1

Risk Interpretation:

0 points = **Low** → No routine prophylaxis

1-2 points = **Intermediate** → Consider prophylaxis; discuss with patient

≥3 points = **High** → Prophylaxis recommended (ASCO)

Note: *ASCO recommends prophylaxis consideration at score ≥2; NCCN categorizes ≥3 as high risk. The Khorana score is validated for ambulatory patients initiating chemotherapy, not for hospitalized patients or those with established VTE*

National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines): Cancer-Associated Venous Thromboembolic Disease. Version 2.2025. Published May 22, 2025. Accessed January 10, 2026. https://www.nccn.org/professionals/physician_gls/pdf/vte.pdf

Case 1 Continued: Risk Stratification

Khorana Score:

- **Cancer site:** Pancreas = +2 points
- **Hemoglobin**: 9.1 g/dL = +1 point; total point 3 points

Assessment: High Risk Category

May consider VTE prophylaxis

Clinical Relevance for This Patient:

- High Khorana score indicates elevated baseline thrombotic risk
- Supports need for prophylactic anticoagulation
- Suggests extended duration therapy will likely be needed given metastatic cancer

Case 1 Continued: Treatment Plan

Which of the following is the appropriate prophylactic plan for this patient:

A. Dalteparin 200 units/kg daily x 30 days, then 150 units/kg daily

B. Apixaban 10 mg BID x 7 days, then 5 mg BID thereafter

C. Rivaroxaban 15 mg BID x 21 days then 20 mg daily

D. Apixaban 5 mg BID, then 2.5 mg daily

Case 1 Continued: Treatment Plan

Which of the following is the appropriate prophylactic plan for this patient:

A. Dalteparin 200 units/kg daily x 30 days, then 150 units/kg daily

B. Apixaban 10 mg BID x 7 days, then 5 mg BID thereafter

C. Rivaroxaban 15 mg BID x 21 days then 10 mg daily

D. Apixaban 2.5 mg daily

Monitoring Parameters:

- Signs and symptoms of bleeding
- Drug interactions
- Renal function in about 3 months

Clinical Case 2: Recurrent VTE

JD is a 63 years old female on pembrolizumab for Stage IV NSCLC presenting with worsening shortness of breath. She was diagnosed with proximal DVT 3 months ago and has been on therapeutic Rivaroxaban since then. A repeat CTPA reveals a segmental pulmonary embolism.

Pertinent Labs:

CrCl: 58 mL/min

Platelets: 225 × 10^9/L

Given the recurrence of VTE while on a DOAC in the setting of active cancer, what is the most appropriate next step in managing new diagnosis?

Case 2 Continued

STEP 1:

- Verify adherence to anticoagulation
- Review for drug to drug interactions
- Confirm appropriate dosing

Adherence confirmed, no major drug interactions identified, dosing appropriate

STEP 2:

- Management Options for Recurrent VTE on Anticoagulation
- Switch drug class (DOAC → LMWH) (ASCO/ITAC/NCCN recommended)

Case 2 Continued

Which of the following is the appropriate treatment plan for this patient:

A. Switch to Dalteparin 200 units/kg daily x 30 days, then 150 units/kg daily

B. Switch to Apixaban 10 mg BID x 7 days, then 5 mg BID thereafter

C. Rivaroxaban 40 mg daily

D. Switch to Warfarin 5 mg daily

Case 2 Continued

Which of the following is the appropriate treatment plan for this patient:

A. **Switch to Dalteparin 200 units/kg daily x 30 days, then 150 units/kg daily**

B. Switch to Apixaban 10 mg BID x 7 days, then 5 mg BID thereafter

C. Rivaroxaban 40 mg daily

D. Switch to Warfarin 5 mg daily

Clinical Pearl:

- Switch to a different class of anticoagulant due to treatment failure
- Recurrent VTE despite appropriate therapeutic anticoagulation is considered treatment failure.

Future Direction in Cancer-Associated VTE Management

- **Factor XI Inhibitors:** offers a new mechanism for targeting intrinsic coagulation pathway without affecting hemostasis
  - Abelacimab is a monoclonal antibody against Factor XI
    - ASTER trial: Phase 3 for cancer associated VTE treatment (ongoing)
- **Enhanced Risk Stratification beyond the Khorana Score: Modified Khorana Scores**
  - **ONKOTEV score**: Adds metastatic disease, prior VTE, vascular compression
  - **PROTECHT:** Adds chemotherapy regimen (platinum, gemcitabine)

Conclusion

- Utilize risk stratification tools like Khorana score to identify high risk ambulatory patients (Score ≥2) who may benefit from VTE prophylaxis.
- DOACs & LMWH are standard of care. Apixaban is preferred for GI/GU malignancies based on CARAVAGGIO safety data
- Treatment should be individualized based on tumor site, renal function, drug interactions and platelet counts.
- Treat for a minimum of 6 months, continue indefinitely in active cancer or if on systemic chemotherapy.
- Recurrent VTE on anticoagulation may warrants switching to a different drug class (DOAC to LMWH), assess for compliance and disease progression.
- Careful consideration in thrombocytopenia (platelet count <50K) and severe renal impairment. LMWH is preferred in these settings.

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Questions?